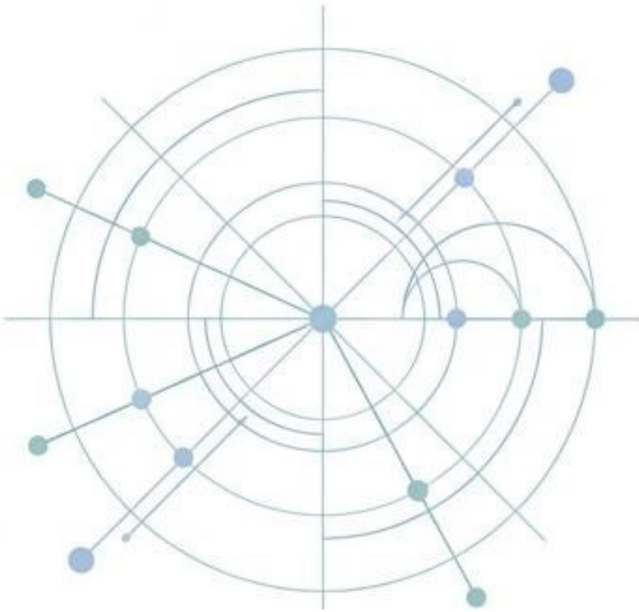
ASEAN
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Data Processing and Preparation

Cleaning, Transformation, and Feature Engineering for Geospatial Analytics



Duration: 30 minutes

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OVERVIEW

Mastering the spatial data preparation workflow



Clean Datasets

Identify and resolve attribute errors and topological inconsistencies.



Handle Missing Data

Deploy advanced imputation and geometric reconstruction techniques.



Transform CRS

Standardize Coordinate Reference Systems for accurate measurement.



Engineer Features

Derive new spatial variables to empower GeoAI and analytics.

OVERVIEW

Poor data quality directly degrades geospatial insights

60–80%

of a data scientist's time is consumed by data preparation and cleaning.

- Operational: Incorrect map generation.
- Analytical: Flawed spatial analysis.
- Algorithmic: Poor machine learning performance.

OVERVIEW

Isolating the three core dimensions of spatial data errors

Attribute Problems

Base tabular errors.

- Mixed spellings
- Missing values
- Duplicate records
- Typographical errors

Geometry Problems

Structural shape errors.

- Missing geometries
- Invalid/broken polygons
- Self-intersections

Spatial Problems

Systemic mapping errors.

- Coordinate Reference System mismatches
- Inconsistent topological boundaries

PIPELINE ARCHITECTURE

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The standard operating procedure for spatial data refinement



Goal: Systematically transition raw geospatial inputs into structurally sound, analysis-ready assets.

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01

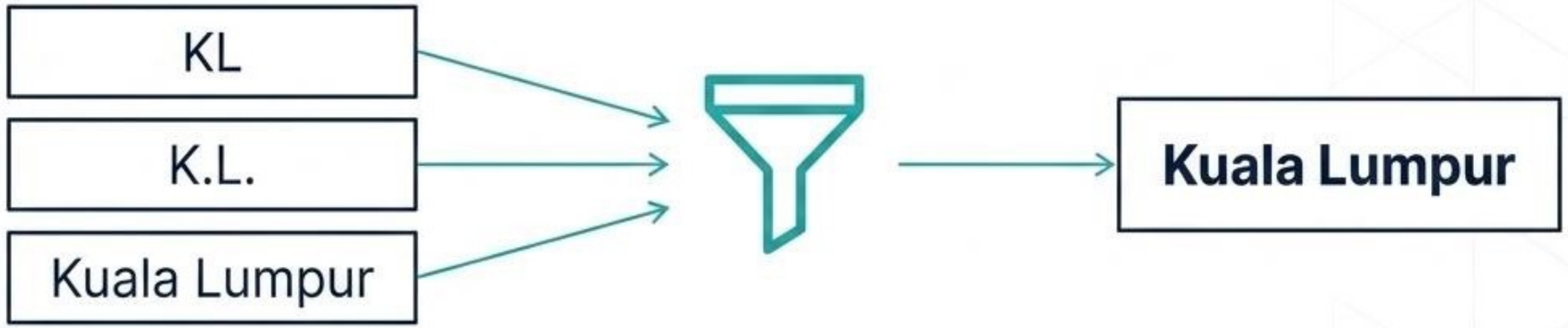
PHASE 01: CLEAN

Attribute & Geometry Cleaning

Resolving duplications, standardizing nomenclature, and enforcing topological integrity.

PHASE 01: ATTRIBUTES

Standardizing textual attributes and eliminating duplicates

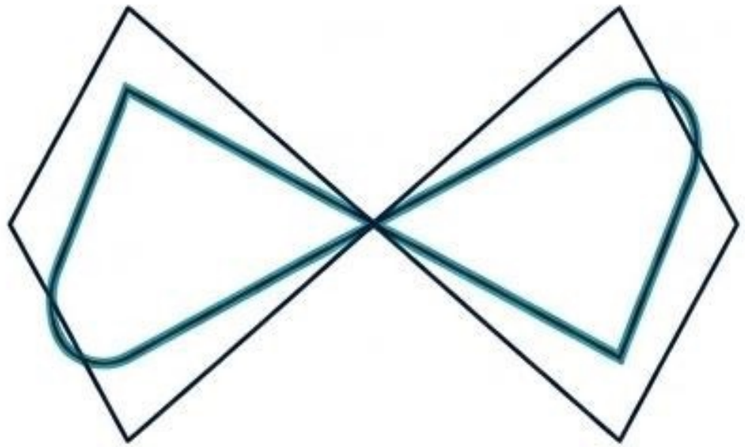


	ID	Name
1	101	School A
2	101	School A

```
Detection: gdf.duplicated()
Removal: gdf.drop_duplicates()
```

Validating topological integrity before spatial operations

Invalid Topology



`Check with gdf.is_valid`

Valid Topology



`Fix with gdf.buffer(0)`

Crucial Prerequisite: Broken geometries must be resolved before initiating spatial joins or distance calculations.

PHASE 01: IMPUTATION

Missing Data Resolution Matrix: Attributes vs. Geometries

Missing Attributes	Missing Geometries
Example: District B has a 'Null' population.	Example: School address exists, but 'No Coordinates'.
Impact: Distorts statistical analysis.	Impact: Renders spatial operations impossible.
Resolution Strategies - Remove (`dropna()`) - Replace (`fillna()`) - Estimate (Mean/Median/Nearest neighbour) - Advanced Spatial Interpolation	Resolution Strategies - Geocoding - GPS field collection - Manual digitization

02

PHASE 02: TRANSFORM

Coordinate Reference Systems

Aligning spatial frames of reference for accurate measurement and integration.

PHASE 02: PROJECTION

Aligning spatial frames of reference for accurate measurement

A CRS defines exactly how real-world locations are mathematically represented on a map.

CRS Type	Primary Use Case
WGS84	Global Positioning Systems (GPS).
UTM	Local mapping and precise distance/area measurement.
Web Mercator	Interactive web mapping platforms.

Why Transform? Required for accurate distance calculations, area measurements, and dataset integration.

```
gdf.to_crs(epsg=3857)
```

03

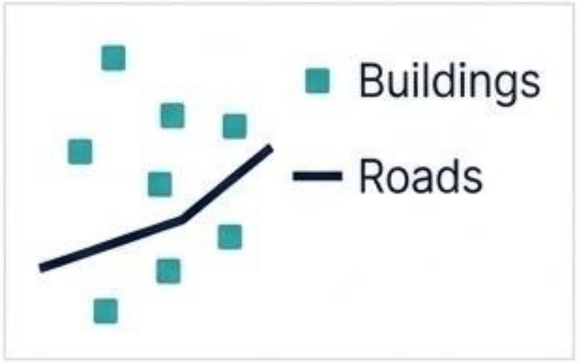
PHASE 03: ENRICH

Feature Engineering

Deriving new spatial variables to optimize analytical and algorithmic performance.

Deriving proximity metrics for advanced site selection

Input Data



Other Variables:

- Population density,
- Elevation,
- Distance to river.

Operation

```
gdf['distance'] =
gdf.distance(roads)
```

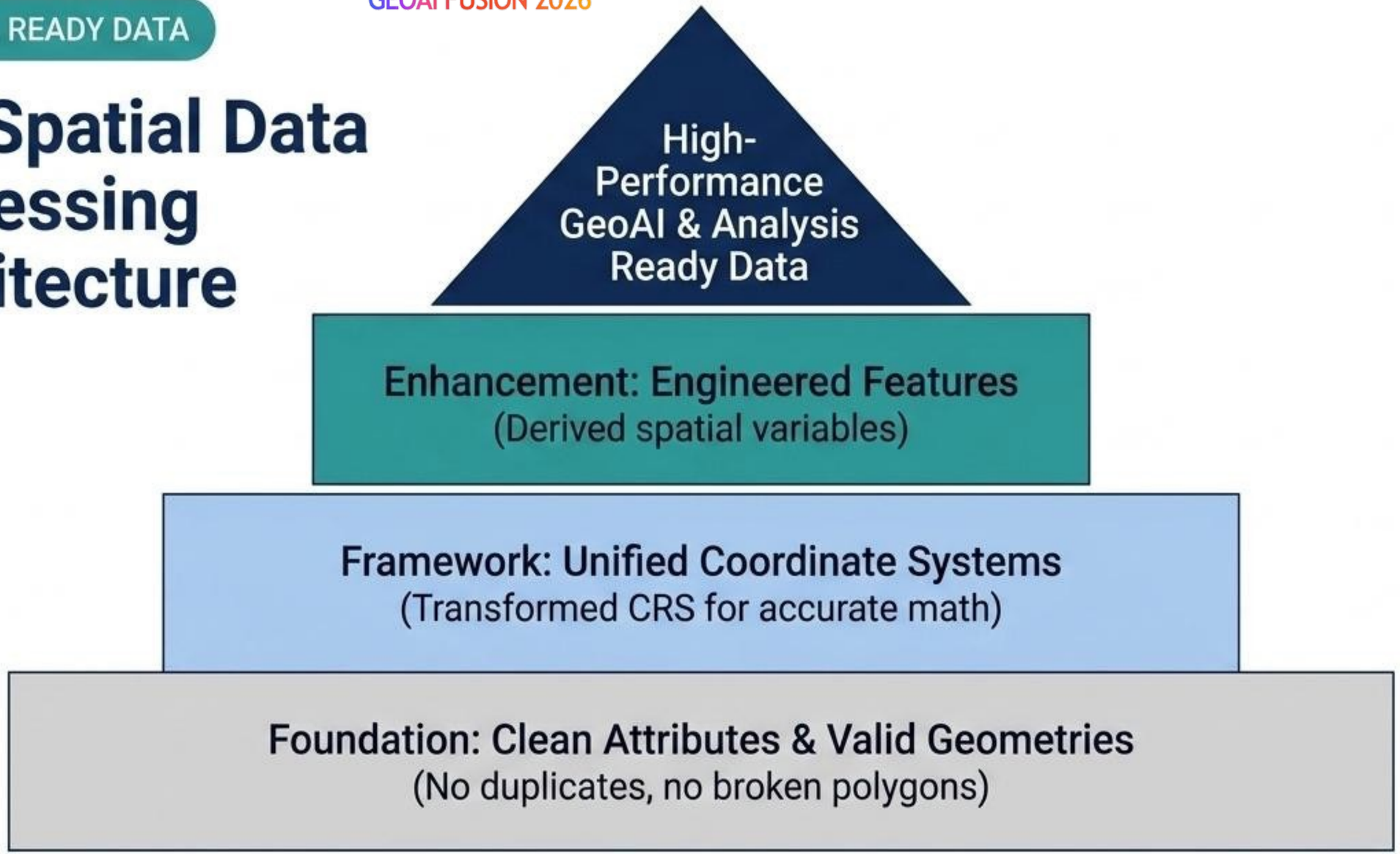
Output Insight

Building	Distance
B1	25m
B2	120m

Applications:
Site selection,
Accessibility analysis,
GeoAI models.

ANALYSIS READY DATA

The Spatial Data Processing Architecture



"High-quality spatial analysis begins with high-quality data."